Arithmetic Monodromy and Middle Convolution of ℓ-adic Perverse Sheaves

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1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics within the mathematical community, we adopt a style of notation inspired by type theory throughout this report, without committing to type-theoretic foundations. The symbols below denote types and/or categories, depending on your philosophical inclination. The notation $X : \mathsf{T}$ means that X is a term of type T or an object of the category T.

Set	Sets
$\mathsf{Set}^{\mathrm{fin}}$	Finite sets
Top	Topological spaces
$\mathsf{Open}(T)$	Open subsets of $T : \mathsf{Top}$
Grp	Groups
$\mathsf{Grp}^{\mathrm{fin}}$	Finite groups
$\mathsf{Grp}^{\mathrm{top}}$	Topological groups
$\mathsf{Grp}^{\mathrm{pro}}$	Profinite groups
Ring	Rings (associative, unital, not necessarily commutative)
$\mathsf{Ring}^{\mathrm{top}}$	Topological rings
Field	Fields
$\mathsf{Field}^{\mathrm{top}}$	Topological fields
Mod_R	Left modules over $R : \mathsf{Ring}$
$\mathsf{Mod}_R^{\mathrm{top}}$	Topological left modules over $R : \mathsf{Ring}^{\mathrm{top}}$
Vect_k	Vector spaces over $k : \mathsf{Field}$
$\mathsf{Vect}_k^{\mathrm{fin}}$	Finite-dimensional vector spaces over $k : \mathsf{Field}$
$\mathsf{Rep}_\Lambda(G)$	Continuous Λ-linear representations of $G : \mathsf{Grp}^{\mathrm{top}}$, where $\Lambda : \mathsf{Ring}^{\mathrm{top}}$
$\mathsf{Rep}_\Lambda^{\mathrm{fin}}(G)$	Continuous Λ-linear representations of $G : \mathsf{Grp}^{\mathrm{top}}$, where $\Lambda : \mathsf{Ring}^{\mathrm{top}}$, whose underlying topological Λ-modules are finitely generated
Sch	Schemes
Sch_S	Schemes over a base scheme $S : \mathsf{Sch}$
Var_k	Varieties over $k : \mathsf{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \mathsf{Field}$, i.e. one-dimensional varieties over k
$\mathsf{Curv}_k^{\mathrm{sm}}$	Smooth curves over $k : \mathsf{Field}$
$\mathsf{Man}_{\mathbb{C}}$	Complex manifolds
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\mathsf{Sh}(X, \mathsf{A})$	Sheaves on $X : \mathsf{Site}$ with values in a category A admitting small limits
$\mathsf{D}^{\mathrm{b}}(\mathsf{A})$	Bounded derived objects of an abelian category A
$X \to Y$	Morphisms from $X : \mathsf{T}$ to $Y : \mathsf{T}$

1.2 Acknowledgements

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1.2.2 Typesetting. The typesetting style of this report is inspired by [Kat88].

1.2.3 Declaration of AI Use. The use of generative AI in the preparation of this report was limited to the following purposes: proofreading; polishing wording, phrasing, grammar, and formatting; helping with LaTeX typesettings; searching for and citing references; suggesting writing advices; answering mathematical questions from the author; translating reading materials in French into English; retypesetting old references with archaic typesettings for ease of reading; generating the TikZ figure Figure 1; providing emotional support for the author; providing fitness advice to help the author stay healthy and energetic while working on his candidacy exam.

1.3 Motivations

1.3.1. *Monodromy groups* are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Roughly speaking, they describe how locally defined data change when transported along a loop. Controlling and understanding monodromy groups has been a central theme in both of these worlds, with significant applications.

1.3.2. The goal of this report is to provide an exposition on monodromy groups as well as ways to study them using methods from perverse sheaves and their middle convolution introduced by Katz [Kat96], especially in the ℓ-adic arithmetic context.

1.3.3. Consider the equation

$$z(u)^2 = u \tag{1.3.3.1}$$

for the unknown holomorphic function $z : V \to \mathbb{C}^\times$, where $V : \mathsf{Open}(\mathbb{C}^\times)$ and $u : V$ is the independent variable. Fix a base point $u_0 : \mathbb{C}^\times$ and choose $z_0 : \mathbb{C}^\times$ with $z_0^2 = u_0$. On a sufficiently small disc V containing u_0, the equation has two holomorphic solution branches $z_+(u) = \sqrt{u}$ and $z_-(u) = -\sqrt{u}$ with $z_+(u_0) = z_0$ and $z_-(u_0) = -z_0$. Move u once around 0 along the loop

$$\gamma : [0,1] \to \mathbb{C}^\times, \qquad t \mapsto u_0 e^{2\pi i t}.$$

Along this loop, the values of the two solution branches vary continuously as $\widetilde{\gamma}_+(t) = z_0 e^{\pi i t}$ and $\widetilde{\gamma}_-(t) = -z_0 e^{\pi i t}$, until they eventually swap places at $t = 1$. In other words, there is a permutation representation of the fundamental group[1] $\pi_1^{\mathrm{top}}(\mathbb{C}^\times, u_0) \cong \mathbb{Z} : \mathsf{Grp}^{\mathrm{top}}$

$$\rho : \pi_1^{\mathrm{top}}(\mathbb{C}^\times, u_0) \to \mathrm{Aut}_{\mathsf{Set}}(\{[z_+]_{u_0}, [z_-]_{u_0}\}), \qquad n \mapsto \left([z_+]_{u_0}\ [z_-]_{u_0}\right)^n.$$

We call this representation the *permutation monodromy representation*, and we call its image, isomorphic to S_2, the *permutation monodromy group*, of the equation (1.3.3.1) at u_0.

[1] with the discrete topology

1.3.4. We reformulate the above with sheaves. Consider the covering

$$f : \mathbb{C}^\times \to \mathbb{C}^\times, \qquad \zeta \mapsto \zeta^2.$$

Let $\mathcal{F} : \mathsf{Sh}(\mathbb{C}^\times, \mathsf{Set})$ be its sheaf of holomorphic sections. Equivalently, this is the sheaf of solutions to the equation (1.3.3.1),

$$\begin{aligned}\mathcal{F}(V) &= \{z : V \to \mathbb{C}^\times \mid z \text{ holomorphic and } (f \circ z)|^V = \mathrm{id}_V\} \\ &= \{z : V \to \mathbb{C}^\times \mid z \text{ holomorphic and } z(u)^2 = u \text{ for every } u : V\}.\end{aligned}$$

As in 1.3.3, on a sufficiently small disc V containing u_0, the sections are precisely the two holomorphic solution branches z_+ and z_-, identifying

$$\mathcal{F}|_V \cong \underline{\{+,-\}}_V.$$

Hence $\mathcal{F}$ is locally constant, with stalk at u_0

$$\mathcal{F}_{u_0} = \{[z_+]_{u_0}, [z_-]_{u_0}\} \cong f^{-1}(u_0)$$

identified by evaluation. However, the sheaf is not constant: $\mathcal{F}(\mathbb{C}^\times) = \emptyset$. A global section would determine a germ at u_0 fixed by all monodromy at u_0, whereas transport once around 0 exchanges the two square-root germs, as demonstrated in 1.3.3. Locally constant (but not necessarily constant) sheaves of sets, such as this, are known as *local systems of sets*.

1.3.5. For a path $\gamma : [0,1] \to \mathbb{C}^\times$ from $u_0 = \gamma(0)$ to $u_1 = \gamma(1)$ and a point $\zeta : f^{-1}(u_0)$ in the fibre, let $\widetilde{\gamma}_\zeta$ be the unique lift through f satisfying $\widetilde{\gamma}_\zeta(0) = \zeta$. We have the *transport map*

$$T_\gamma : \mathcal{F}_{u_0} \xrightarrow{\sim} \mathcal{F}_{u_1}, \qquad \zeta \longmapsto \widetilde{\gamma}_\zeta(1).$$

with the evaluation identifications $\mathcal{F}_u \cong f^{-1}(u)$. Thus we obtain a *transport functor*[2]

$$T : \Pi_1(\mathbb{C}^\times) \longrightarrow \mathsf{Set}, \qquad u \longmapsto \mathcal{F}_u, \qquad [\gamma] \longmapsto T_\gamma.$$

Restricting transport to loops at u_0 gives the permutation monodromy representation of $\mathcal{F}$ at u_0, the same permutation monodromy representation as in 1.3.3

$$\rho_{\mathcal{F},u_0} : \pi_1^{\mathrm{top}}(\mathbb{C}^\times, u_0) \to \mathrm{Aut}_{\mathsf{Set}}(\mathcal{F}_{u_0}), \qquad [\gamma] \mapsto T_\gamma.$$

1.3.6. In § 2, we will see how to upgrade permutation monodromy representations into linear monodromy representations, which we will just call monodromy representations. Moreover, one can use this machinery to talk about monodromy in the ℓ-adic arithmetic context, by replacing π_1^{top} with the étale fundamental group $\pi_1^{\text{ét}}$. When $U = \mathrm{Spec}(k)$ for some k : Field, with a choice of separable closure k^{sep}, then $\pi_1(U, u_0) = \mathrm{Gal}(k^{\mathrm{sep}}/k)$, where $u_0 = \mathrm{Spec}(k^{\mathrm{sep}})$, is the absolute Galois group of k. Hence, a monodromy representation at u_0 in this case degenerates into a Galois representation, a ubiquitous arithmetic object.

[2]Here $\Pi_1(X)$ denotes the fundamental groupoid of some X : Top, that is, the category whose objects are points of X and whose morphisms are homotopy classes of paths relative to their endpoints, with concatenation as composition.

2 Local Systems and Lisse Sheaves

2.1 Complex Local Systems

2.1.1. We have looked at permutation monodromy, but permutation actions do not capture all the situations we wish to study, since a lot of times in nature we encounter actions on vector spaces instead of sets. For example, instead of an algebraic equation like the one in 1.3.3, we consider a differential equation, which is known as *Gauss's hypergeometric equation*

$$u(1-u)\frac{\mathrm{d}^2 z}{\mathrm{d}u^2} + \big(c-(a+b+1)u\big)\frac{\mathrm{d}z}{\mathrm{d}u} - abz = 0 \tag{2.1.1.1}$$

where $a, b, c : \mathbb{C}$ are fixed parameters and $z : V \to \mathbb{C}$ is an unknown holomorphic function, with $V : \mathsf{Open}(U)$, $U = \widehat{\mathbb{C}} \setminus \{0, 1, \infty\}$, and independent variable $u : V$. We then define the sheaf of solutions to this equation $\mathcal{F} : \mathsf{Sh}(U, \mathsf{Vect}_{\mathbb{C}})$

$$\mathcal{F}(V) = \{z : V \to \mathbb{C} \mid z \text{ holomorphic and satisfies (2.1.1.1) on } V\}.$$

This is a sheaf of vector spaces since (2.1.1.1) is a homogeneous linear differential equation. For each $u_0 : U$, the existence-and-uniqueness theorem [HTT08, Theorem 4.3.1] gives a sufficiently small open disc $V : \mathsf{Open}(U)$ containing u_0 and an isomorphism of sheaves

$$\mathcal{F}|_V \xrightarrow{\sim} \underline{\mathbb{C}}^2_V$$

Thus $\mathcal{F}$ is locally constant of rank 2. We call such a sheaf a *complex local system of rank* 2.

2.1.2 Definition. (Complex Local System) Let $U : \mathsf{Man}_{\mathbb{C}}$ be connected. A *complex local system* on U is a locally constant sheaf $\mathcal{F} : \mathsf{Sh}(U, \mathsf{Vect}_{\mathbb{C}})$ with finite-dimensional stalks. Explicitly, every $u : U$ has an open neighbourhood $V : \mathsf{Open}(U)$ and an isomorphism

$$\varphi_V : \mathcal{F}|_V \xrightarrow{\sim} \underline{\mathbb{C}}^n_V$$

called a *local trivialization* at u, for some $n : \mathbb{N}$ called the *rank* of $\mathcal{F}$. Denote by $\mathsf{LocSys}_{\mathbb{C}}(U)$ the (symmetric monoidal $\mathbb{C}$-linear) category of complex local systems on U.

2.1.3 Definition. (Transport) Let $\mathcal{F}$ be a complex local system of rank n on a connected $U : \mathsf{Man}_{\mathbb{C}}$. We define the *transport functor* using analytic continuation (see Appendix A.1.1)

$$T : \Pi_1(U) \to \mathsf{Vect}^{\mathrm{fin}}_{\mathbb{C}}, \qquad u \mapsto \mathcal{F}_u, \qquad [\gamma] \mapsto T_\gamma.$$

where the *transport map* $T_\gamma : \mathcal{F}_{\gamma(0)} \xrightarrow{\sim} \mathcal{F}_{\gamma(1)}$ for a path $\gamma : [0,1] \to U$ is defined as follows. By compactness of closed intervals, we can choose a subdivision $0 = t_0 < \cdots < t_N = 1$ and open sets $V_i : \mathsf{Open}(U)$ containing $\gamma([t_{i-1}, t_i])$, with local trivializations

$$\varphi_i : \mathcal{F}|_{V_i} \xrightarrow{\sim} \underline{\mathbb{C}}^n_{V_i}.$$

Put $v_i = \gamma(t_i)$ for each $i = 0, \ldots, N$, then the trivializations induce stalk isomorphisms $\varphi_{i,v_i} : \mathcal{F}_{v_i} \xrightarrow{\sim} \mathbb{C}^n$. The transport map along γ is then defined as the composite

$$T_\gamma = \big(\varphi^{-1}_{N,v_N} \circ \varphi_{N,v_{N-1}}\big) \circ \cdots \circ \big(\varphi^{-1}_{1,v_1} \circ \varphi_{1,v_0}\big) : \mathcal{F}_{\gamma(0)} \xrightarrow{\sim} \mathcal{F}_{\gamma(1)}.$$

which is independent of the subdivision and trivializations and depends only on the homotopy class of γ relative to its endpoints, by monodromy theorem [Sch11, Lemma 7.2.7].

2.1.4 Definition. (Monodromy) Let $\mathcal{F}$ be a complex local system of rank n on a connected $U : \mathsf{Man}_{\mathbb{C}}$, and fix a base point $u_0 : U$. Restricting its transport functor to loops at u_0 gives the *monodromy representation* of $\mathcal{F}$ at u_0

$$\rho_{\mathcal{F},u_0} : \pi_1^{\mathrm{top}}(U, u_0) \to \mathrm{Aut}_{\mathsf{Vect}_{\mathbb{C}}}(\mathcal{F}_{u_0}), \qquad [\gamma] \mapsto T_\gamma,$$

whose image is the *monodromy group* of $\mathcal{F}$ at u_0.

2.1.5. Like in 1.3.4, $\mathcal{F}$ is also a sheaf of sections of a cover. The *espace étalé* of U is

$$\text{Ét}(U) := \coprod_{u \in U} \mathcal{F}_u : \mathsf{Man}_{\mathbb{C}}$$

with topology generated by sets $\widetilde{s}(V) = \{(u, s_u) : u : V\}$ for $V : \mathsf{Open}(U)$ and $s : \mathcal{F}(V)$. Each basic open $\widetilde{s}(V)$ is identified with V by $(u, s_u) \mapsto u$, so the complex charts of U pull back to a compatible complex atlas on $\text{Ét}(U)$.[3] Its associated covering map

$$f : \text{Ét}(U) \to U, \qquad (u, \xi) \mapsto u$$

is the analogue of the double cover in 1.3.4, and its sheaf of sections recovers $\mathcal{F}$. Analytic continuation on U is equivalent to path lifting to the espace étalé, so the transport can also be defined equivalently via path lifting, such as in 1.3.5.

2.1.6. In the other direction, if we are given a representation

$$\rho : \pi_1^{\mathrm{top}}(U, u_0) \to \mathrm{Aut}_{\mathsf{Vect}_{\mathbb{C}}}(W)$$

for some connected $U : \mathsf{Man}_{\mathbb{C}}$, base point $u_0 : U$ and $W : \mathsf{Vect}_{\mathbb{C}}^{\mathrm{fin}}$ of dimension n, we can recover a local system $\mathcal{F}$ of rank n on U whose monodromy representation is ρ by the following way. Pick a universal cover $q : \widetilde{U} \to U$ and a basepoint lift $\widetilde{u_0} : q^{-1}(u_0)$. Define a right $\pi_1^{\mathrm{top}}(U, u_0)$ action on $\widetilde{U} \times W_{\mathrm{disc}}$, where W_{disc} has the discrete topology, by

$$(\widetilde{u}, w) \cdot g = (\widetilde{u} \cdot g, \rho(g)^{-1} w).$$

where $\widetilde{u} \cdot g$ is the action by the deck transformation determined by $\widetilde{u_0}$. Form the orbit space

$$E_\rho = (\widetilde{U} \times W_{\mathrm{disc}}) / \pi_1^{\mathrm{top}}(U, u_0)$$

We note that E_ρ inherits a complex structure induced by its associated covering

$$f_\rho : E_\rho \to U, \qquad [\widetilde{u}, w] \mapsto q(\widetilde{u})$$

for which it is a local biholomorphism i.e. étale cover. These recover the espace étalé of our desired local system and its associated cover. Therefore, we can take its sheaf of sections $\mathcal{F}_\rho$ to obtain our desired local system, where for each $V : \mathsf{Open}(U)$

$$\begin{aligned} \mathcal{F}_\rho(V) &= \{s : V \to E_\rho \mid s \text{ holomorphic}^4 \text{ and } (f_\rho \circ s)|^V = \mathrm{id}_V\} \\ &\cong \left\{ a : q^{-1}(V) \to W_{\mathrm{disc}} \;\middle|\; \begin{array}{l} a \text{ locally constant,} \\ a(\widetilde{u} \cdot g) = \rho(g)^{-1} a(\widetilde{u}) \end{array} \right\}. \end{aligned}$$

[3] Throughout this report, complex manifolds are not required to be second-countable. In particular, the *espace étalé* of a positive-rank complex local system has uncountable discrete fibres and hence is not second-countable.

where the last identification is given by $s(q(\widetilde{u})) = [\widetilde{u}, a(\widetilde{u})]$ for every $\widetilde{u} : q^{-1}(V)$.

2.1.7 Definition. (Linear Étale Cover) Let $U : \mathsf{Man}_{\mathbb{C}}$ be connected. A *linear étale cover* of U is a complex manifold $E : \mathsf{Man}_{\mathbb{C}}$ equipped with a holomorphic covering map $f : E \to U$ and a finite-dimensional complex vector-space structure on each fibre, such that every $u : U$ has an open neighbourhood $V : \mathsf{Open}(U)$ and a biholomorphism over V

$$\varphi_V : f^{-1}(V) \xrightarrow{\sim} V \times W_{\mathrm{disc}}$$

which is linear on each fibre, for some $W : \mathsf{Vect}_{\mathbb{C}}^{\mathrm{fin}}$ with W_{disc} given the discrete topology. Denote by $\acute{\mathsf{Et}}_{\mathbb{C}}^{\mathrm{lin}}(U)$ the category of these covers, with fibrewise linear holomorphic maps over U as morphisms.

2.1.8. For each $U : \mathsf{Man}_{\mathbb{C}}$ and $u_0 : U$, there are equivalences of categories [Mac90, § 7.2]

$$\acute{\mathsf{Et}}_{\mathbb{C}}^{\mathrm{lin}}(U) \simeq \mathsf{LocSys}_{\mathbb{C}}(U) \simeq^{\otimes} \mathsf{Rep}_{\mathbb{C}}^{\mathrm{fin}}\big(\pi_1^{\mathrm{top}}(U, u_0)\big).$$

Moreover, the last equivalence is of symmetric monoidal $\mathbb{C}$-linear categories.

2.1.9 Example. A rich source of local systems is smooth and proper families. Let

$$f : E \to U$$

be a smooth proper map between complex manifolds. The fibres $E_{u_0} := f^{-1}(u_0)$ at $u_0 : U$ are compact complex manifolds. The higher direct image sheaves

$$\mathbf{R}^q f_* \underline{\mathbb{C}}$$

are complex local systems for each $q : \mathbb{N}$ by Ehresmann's fibration theorem [Voi02, Theorem 9.3]. Explicitly, at $u_0 : U$, there exists open neighborhood $V : \mathsf{Open}(U)$ of u_0 with

$$\mathbf{R}^q f_* \underline{\mathbb{C}}|_V \cong \underline{\mathrm{H}^q(E_{u_0}, \underline{\mathbb{C}})}_V$$

In particular on the level of stalks $\big(\mathbf{R}^q f_* \underline{\mathbb{C}}\big)_{u_0} \cong \mathrm{H}^q(E_{u_0}, \underline{\mathbb{C}})$.

2.2 ℓ-adic Local Systems

2.2.1. We now develop the ℓ-adic analogues of the constructions in § 2.1.

2.2.2. In number theory, ℓ-adic Galois representations are ubiquitous and encode a lot of intricate arithmetic information. Let $E : \mathsf{Curv}_k^{\mathrm{sm}}$ be an elliptic curve over a field $k : \mathsf{Field}$, recall that its Tate module at a prime $\ell : \mathbb{N}$ is the $\mathbb{Z}_\ell[\Gamma_k]$-module

$$\mathrm{T}_\ell(E) = \varprojlim E[\ell^n]$$

[4] Continuity alone suffices: since f_ρ is a local biholomorphism, every continuous section is locally a holomorphic inverse of f_ρ, and hence is holomorphic.

where $\Gamma_k = \mathrm{Gal}(k^{\mathrm{sep}}/k)$. In other words, it is an (continuous) ℓ-adic Galois representation

$$\rho_{E,\ell} : \Gamma_k \to \mathrm{Aut}_{\mathbb{Z}_\ell}(\mathrm{T}_\ell(E))$$

Theorems of Faltings and Zarhin–Mori show that the rationalized Tate module

$$\mathrm{V}_\ell(E) := \mathrm{T}_\ell(E) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

as a $\mathbb{Q}_\ell[\Gamma_k]$-module determines the elliptic curve up to k-isogeny, assuming k is finitely generated over its prime field [Fal83, Zar76, Mor78]. In addition, the Tate module also contains information about reduction types (c.f. Néron–Ogg–Shafarevich criterion [Sil09, Theorem VII.7.1] and Serre–Tate theorem [ST68, Theorem 1]), endomorphisms (c.f. Tate's theorem [Tat66, Main Theorem]), point counts over finite fields (c.f. Grothendieck–Lefschetz trace formula [Mil13, Theorem 29.4]), and many more.

2.2.3 Notation. We use the following notation:

$k : \mathsf{Field}$	A field
$U : \mathsf{Sch}_{\mathrm{Spec}(k)}$	A Noetherian scheme over k
$\ell : \mathbb{N}$	A prime with $\ell \neq \mathrm{char}(k)$
$U_{\text{ét}} : \mathsf{Site}$	The small étale site of U

2.2.4 Definition. (Mod-ℓ^n Sheaf) For each $n : \mathbb{N}$, a *mod-ℓ^n sheaf* on U is a sheaf of $\mathbb{Z}/\ell^n\mathbb{Z}$-modules on $U_{\text{ét}}$. Denote their (symmetric monoidal $\mathbb{Z}/\ell^n\mathbb{Z}$-linear) category as

$$\mathsf{Sh}(U, \mathbb{Z}/\ell^n\mathbb{Z}) := \mathsf{Sh}(U_{\text{ét}}, \mathsf{Mod}_{\mathbb{Z}/\ell^n\mathbb{Z}}).$$

A mod-ℓ^n sheaf is *lisse* if it is locally constant with finite geometric stalks. We denote by $\mathsf{LocSys}_{\mathbb{Z}/\ell^n\mathbb{Z}}(U)$ the (symmetric monoidal $\mathbb{Z}/\ell^n\mathbb{Z}$-linear) full subcategory of lisse sheaves.

2.2.5 Definition. (ℓ-adic Sheaf) An *(integral) ℓ-adic sheaf* on U is a family $\mathcal{F}$ of sheaves

$$\mathcal{F}_n : \mathsf{Sh}(U, \mathbb{Z}/\ell^n\mathbb{Z})$$

equipped with maps $r_n : \mathcal{F}_{n+1} \to \mathcal{F}_n$ in $\mathsf{Sh}(U, \mathbb{Z}/\ell^{n+1}\mathbb{Z})$, such that

$$\overline{r_n} : \mathcal{F}_{n+1}/\ell^n\mathcal{F}_{n+1} \xrightarrow{\sim} \mathcal{F}_n$$

factoring through r_n is an isomorphism in $\mathsf{Sh}(U, \mathbb{Z}/\ell^{n+1}\mathbb{Z})$, for each $n : \mathbb{N}$. Moreover, we say that $\mathcal{F}$ is *lisse* if each $\mathcal{F}_n$ is lisse. Denote by $\mathsf{Sh}(U, \mathbb{Z}_\ell)$ the (symmetric monoidal $\mathbb{Z}_\ell$-linear) category of ℓ-adic sheaves on U, and $\mathsf{LocSys}_{\mathbb{Z}_\ell}(U)$ the (symmetric monoidal $\mathbb{Z}_\ell$-linear) full subcategory of lisse sheaves.

2.2.6 Definition. (Rational ℓ-adic Sheaf) Define the category of *rational ℓ-adic sheaves* on U, denoted $\mathsf{Sh}(U, \mathbb{Q}_\ell)$, as the (symmetric monoidal $\mathbb{Q}_\ell$-linear) category where the objects are the same as $\mathsf{Sh}(U, \mathbb{Z}_\ell)$ but the morphisms between two objects $\mathcal{F}, \mathcal{G} : \mathsf{Sh}(U, \mathbb{Q}_\ell)$ are

$$\mathrm{Hom}_{\mathsf{Sh}(U,\mathbb{Q}_\ell)}(\mathcal{F}, \mathcal{G}) := \mathrm{Hom}_{\mathsf{Sh}(U,\mathbb{Z}_\ell)}(\mathcal{F}, \mathcal{G}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

considered as tensoring of $\mathbb{Z}_\ell$-modules. An object of this category is a rational ℓ-adic sheaf on U. There is a canonical functor called *rationalization*

$$(-)_{\mathbb{Q}_\ell} = (-) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell : \mathsf{Sh}(U, \mathbb{Z}_\ell) \to \mathsf{Sh}(U, \mathbb{Q}_\ell), \qquad \mathcal{F} \mapsto \mathcal{F}, \qquad f \mapsto f \otimes 1.$$

A rational ℓ-adic sheaf $\mathcal{F}$ on U is *lisse* if there exists a lisse integral ℓ-adic sheaf $\mathcal{L}$ and an isomorphism $\mathcal{F} \cong \mathcal{L} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. Denote by $\mathsf{LocSys}_{\mathbb{Q}_\ell}(U)$ the (symmetric monoidal $\mathbb{Q}_\ell$-linear) full subcategory of lisse sheaves of $\mathsf{Sh}(U, \mathbb{Q}_\ell)$.

2.2.7. Many common sheaf constructions for integral ℓ-adic sheaves can be taken level-wise such as pullbacks, direct sums, and tensor products, but unfortunately there are also many common sheaf constructions that require corrections, since an operation could produce a family $\mathcal{A}$ that does not satisfy

$$\mathcal{A}_{n+1} / \ell^n \mathcal{A}_{n+1} \cong \mathcal{A}_n$$

Whenever such a representative exists, we replace the family with an object that does satisfy this condition and is isomorphic to it in the Artin–Rees category [SGA5, Exp. V]. Explicitly, we replace $\mathcal{A}$ with $\mathcal{A}'$ where

$$\mathcal{A}'_n = \mathcal{B}_N / \ell^n \mathcal{B}_N, \qquad \text{where} \qquad \mathcal{B}_n := \bigcap_{m \geq n} \operatorname{Im}(\mathcal{A}_m \longrightarrow \mathcal{A}_n),$$

for each $n : \mathbb{N}$, with $N : \mathbb{N}$ sufficiently large, dependent on n. Constructions that require such corrections include kernels, images, sheaf Hom, and duals.

2.2.8. Some constructions of integral ℓ-adic sheaves require taking inverse limits. For an integral ℓ-adic sheaf $\mathcal{F}$, these include:

Name	**Formula**	**Assumptions**
Geometric stalks	$\mathcal{F}_{u_0} := \varprojlim_n (\mathcal{F}_n)_{u_0}$	$u_0 : \operatorname{Spec}(\Omega) \to U$ $\Omega : \mathsf{Field}$, separably closed
Sections	$\Gamma(V, \mathcal{F}) := \varprojlim_n \Gamma(V, \mathcal{F}_n)$	$V : U_{\text{ét}}$
Sheaf cohomology	$\mathrm{H}^q(U, \mathcal{F}) := \varprojlim_n \mathrm{H}^q(U_{\text{ét}}, \mathcal{F}_n)$	$q : \mathbb{N}$
Morphism modules	$\operatorname{Hom}(\mathcal{F}, \mathcal{G}) \cong \varprojlim_n \operatorname{Hom}_{\mathsf{Sh}(U, \mathbb{Z}/\ell^n\mathbb{Z})}(\mathcal{F}_n, \mathcal{G}_n)$	$\mathcal{G} : \mathsf{Sh}(U, \mathbb{Z}_\ell)$

2.2.9. Rational ℓ-adic sheaves require corrections by rationalizing the integral constructions. For a rational ℓ-adic sheaf $\mathcal{F}_{\mathbb{Q}_\ell} = \mathcal{F} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ with $\mathcal{F}$ an integral ℓ-adic sheaf, constructions of $\mathcal{F}_{\mathbb{Q}_\ell}$ are usually done by first applying the corresponding (corrected) constructions for $\mathcal{F}$, and then rationalizing. For example:

Name	Formula	Assumptions
Geometric stalks	$(\mathcal{F}_{\mathbb{Q}_\ell})_{u_0} := \mathcal{F}_{u_0} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$u_0 : \mathrm{Spec}(\Omega) \to U$ $\Omega : \mathsf{Field}$, separably closed
Sections	$\Gamma(V, \mathcal{F}_{\mathbb{Q}_\ell}) := \Gamma(V, \mathcal{F}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$V : U_{\text{ét}}$
Sheaf cohomology	$\mathrm{H}^q(U, \mathcal{F}_{\mathbb{Q}_\ell}) := \mathrm{H}^q(U, \mathcal{F}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$q : \mathbb{N}$
Kernel	$\mathrm{Ker}(f \otimes 1) := (\mathrm{Ker}\, f) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$f : \mathcal{F} \to \mathcal{G}$ in $\mathsf{Sh}(U, \mathbb{Z}_\ell)$
Image	$\mathrm{Im}(f \otimes 1) := (\mathrm{Im}\, f) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$f : \mathcal{F} \to \mathcal{G}$ in $\mathsf{Sh}(U, \mathbb{Z}_\ell)$
Cokernel	$\mathrm{Coker}(f \otimes 1) := (\mathrm{Coker}\, f) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$f : \mathcal{F} \to \mathcal{G}$ in $\mathsf{Sh}(U, \mathbb{Z}_\ell)$
Coimage	$\mathrm{Coim}(f \otimes 1) := (\mathrm{Coim}\, f) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$	$f : \mathcal{F} \to \mathcal{G}$ in $\mathsf{Sh}(U, \mathbb{Z}_\ell)$
Pullback	$g^*(\mathcal{F}_{\mathbb{Q}_\ell}) := (g^*\mathcal{F})_{\mathbb{Q}_\ell}$	$g : V \to U$ $V : \mathsf{Sch}_{\mathrm{Spec}(k)}$, Noetherian
Direct sum	$\mathcal{F}_{\mathbb{Q}_\ell} \oplus \mathcal{G}_{\mathbb{Q}_\ell} := (\mathcal{F} \oplus \mathcal{G})_{\mathbb{Q}_\ell}$	$\mathcal{G} : \mathsf{Sh}(U, \mathbb{Z}_\ell)$
Tensor product	$\mathcal{F}_{\mathbb{Q}_\ell} \otimes_{\mathbb{Q}_\ell} \mathcal{G}_{\mathbb{Q}_\ell} := (\mathcal{F} \otimes_{\mathbb{Z}_\ell} \mathcal{G})_{\mathbb{Q}_\ell}$	$\mathcal{G} : \mathsf{Sh}(U, \mathbb{Z}_\ell)$
Sheaf Hom	$\mathcal{H}om_{\mathbb{Q}_\ell}(\mathcal{F}_{\mathbb{Q}_\ell}, \mathcal{G}_{\mathbb{Q}_\ell}) := (\mathcal{H}om_{\mathbb{Z}_\ell}(\mathcal{F}, \mathcal{G}))_{\mathbb{Q}_\ell}$	$\mathcal{G} : \mathsf{Sh}(U, \mathbb{Z}_\ell)$
Dual	$(\mathcal{F}_{\mathbb{Q}_\ell})^\vee := (\mathcal{F}^\vee)_{\mathbb{Q}_\ell}$	$\mathcal{F}^\vee := \mathcal{H}om_{\mathbb{Z}_\ell}(\mathcal{F}, \mathbb{Z}_\ell)$

2.2.10. We note that Bhatt and Scholze [BS15] have developed the framework of *pro-étale sites*, allowing ℓ-adic sheaves to be considered as bona fide sheaves on the pro-étale site $U_{\text{pro-ét}}$, which removes the clunkiness we had to deal with in 2.2.7, 2.2.8, and 2.2.9.

2.2.11. Recall the étale fundamental group of a connected U at a geometric base point $u_0 : \mathrm{Spec}(\Omega) \to U$ is the profinite group defined by the inverse limit

$$\pi_1^{\text{ét}}(U, u_0) := \varprojlim_{(X, x_0) \to (U, u_0)} \mathrm{Aut}_U(X) : \mathsf{Grp}^{\text{pro}}$$

where the limit is taken over connected finite étale Galois covers $X \to U$ with x_0 a lift of u_0, and the transition map $\varphi : \mathrm{Aut}_U(Y) \to \mathrm{Aut}_U(X)$ for a pointed map $h : (Y, y_0) \to (X, x_0)$ between conneced finite étale Galois covers is given by $h \circ \sigma = \varphi(\sigma) \circ h$, for $\sigma : \mathrm{Aut}_U(Y)$.

2.2.12. Alternatively, one can construct the étale fundamental group via the *fibre functor*

$$\omega_{u_0} : \acute{\mathsf{Et}}^{\text{fin}}(U) \to \mathsf{Set}^{\text{fin}}, \qquad E \mapsto \mathrm{Hom}_U(\mathrm{Spec}(\Omega), E)$$

where $\acute{\mathsf{Et}}^{\text{fin}}(U)$ is the category of finite étale covers $X \to U$ with morphisms over U. We can then construct the étale fundamental group as

$$\pi_1^{\text{ét}}(U, u_0) = \mathrm{Aut}(\omega_{u_0}) : \mathsf{Grp}^{\text{pro}}$$

with topology given by the subspace topology induced by

$$\pi_1^{\text{ét}}(U, u_0) \hookrightarrow \prod_{E : \text{Ét}^{\text{fin}}(U)} \text{Sym}(\omega_{u_0}(E))$$

where each symmetric group is given the discrete topology.

2.2.13. Just as in 2.1.8, for connected U, a geometric base point $u_0 : \text{Spec}(\Omega) \to U$, and a coefficient ring $\Lambda : \text{Ring}^{\text{top}}$, being one of $\mathbb{Z}_\ell$, $\mathbb{Q}_\ell$, or $\mathbb{Z}/\ell^n\mathbb{Z}$ for some $n : \mathbb{N}$, we have a symmetric monoidal Λ-linear equivalence of categories

$$\text{LocSys}_\Lambda(U) \simeq^{\otimes} \text{Rep}_\Lambda^{\text{fin}}(\pi_1^{\text{ét}}(U, u_0))$$

We first construct this for $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$ for $n : \mathbb{N}$, and then the integral and rational ℓ-adic cases follow from assembling each n and then rationalizing.

2.2.14. Let $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$ for $n : \mathbb{N}$, and $\mathcal{F} : \text{LocSys}_\Lambda(U)$. Let $E \to U$ be the finite étale cover representing the underlying sheaf of sets of $\mathcal{F}$, we have a (continuous) representation

$$\rho_{\mathcal{F}} : \pi_1^{\text{ét}}(U, u_0) \to \text{Aut}_{\text{Mod}_\Lambda}(\mathcal{F}_{u_0})$$

with the action by $\sigma : \pi_1^{\text{ét}}(U, u_0)$ induced by

$$\sigma_E : \omega_{u_0}(E) \to \omega_{u_0}(E)$$

under the canonical identification

$$\mathcal{F}_{u_0} = \varinjlim_{\substack{q : V \to U \text{ étale} \\ v : \text{Spec}(\Omega) \to V \text{ geo. pt.} \\ q \circ v = u_0}} \mathcal{F}(V) \xrightarrow{\sim} \omega_{u_0}(E), \qquad [(V, v), s] \mapsto s \circ v$$

where $(s : V \to E) : \text{Hom}_U(V, E) = \mathcal{F}(V)$ is representing a section.

2.2.15. Conversely, given a (continuous) representation in $\text{Rep}_\Lambda^{\text{fin}}(\pi_1^{\text{ét}}(U, u_0))$

$$\rho : \pi_1^{\text{ét}}(U, u_0) \to \text{Aut}_{\text{Mod}_\Lambda}(M)$$

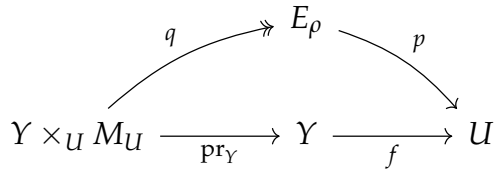
Let $N = \text{Ker}(\rho)$, which is open by continuity, so $G = \pi_1^{\text{ét}}(U, u_0)/N$ is finite and ρ factorizes

$$\pi_1^{\text{ét}}(U, u_0) \twoheadrightarrow G \xrightarrow{\bar{\rho}} \text{Aut}_{\text{Mod}_\Lambda}(M).$$

By Grothendieck's Galois theory (see Appendix A.3), the $\pi_1^{\text{ét}}(U, u_0)$-set G corresponds to a finite étale cover $f : Y \to U$ with $\omega_{u_0}(Y) \cong G$ as $\pi_1^{\text{ét}}(U, u_0)$-sets. By the identification we have a free transitive right G-action on $\omega_{u_0}(Y)$ equivariant with the $\pi_1^{\text{ét}}(U, u_0)$-action. By Grothendieck's Galois theory, this corresponds to a deck transformation of Y transitive and free on each geometric fibre, which makes Y an étale G-torsor. By finiteness of M, set $M_U := (\coprod_{m : M} U \to U) : \text{Sch}_U$ and form the space called the *contracted product*

$$E_\rho := Y \times^G M := (Y \times_U M_U)/G \cong \left(\coprod_{m : M} Y \right) \Big/ G$$

where G acts by $(y, m) \cdot g = (yg, \overline{\rho}(g)^{-1}m)$. There is a unique map $p : E_\rho \to U$ factoring through the G-invariant composite map



where q is the quotient map. Thus we found a finite étale cover

$$(E_\rho \xrightarrow{p} U) : \acute{\mathrm{E}}\mathrm{t}^{\mathrm{fin}}(U)$$

Moreover, the module operations on M are G-equivariant and descend to addition, zero, and scalar-multiplication morphisms over U, making E_ρ a Λ-module scheme over U. By taking its sheaf of sections

$$\mathcal{F}_\rho(V) := \mathrm{Hom}_U(V, E_\rho) = \{s : V \to E_\rho \mid p \circ s = q\} : \mathsf{Mod}_\Lambda,$$

for $q : V \to U$ étale, we obtain our desired lisse sheaf $\mathcal{F}_\rho : \mathsf{LocSys}_\Lambda(U)$. To see that $\mathcal{F}_\rho$ is étale-locally constant, note that pulling back by $f : Y \to U$ gives

$$\mathcal{F}_\rho|_Y \cong f^*\mathcal{F}_\rho \cong \mathrm{Hom}_Y(-, Y \times_U E_\rho) \cong \mathrm{Hom}_Y\left(-, \coprod_{m\,:\,M} Y\right) \cong \underline{M}_Y.$$

2.2.16. Let $\Lambda = \mathbb{Z}_\ell$. Given a continuous representation

$$\rho : \pi_1^{\text{ét}}(U, u_0) \to \mathrm{Aut}_{\mathbb{Z}_\ell}(M)$$

on a finitely generated $\mathbb{Z}_\ell$-module $M : \mathsf{Mod}_\Lambda^{\mathrm{top}}$ with its ℓ-adic topology, apply 2.2.15 to the induced representations ρ_n on $M_n := M/\ell^n M$ for $n \geq 1$, obtaining sheaves $\mathcal{F}_n$. The reductions $M_{n+1} \twoheadrightarrow M_n$ are equivariant, and, by Grothendieck's Galois theory, they induce transition maps $r_n : \mathcal{F}_{n+1} \to \mathcal{F}_n$ satisfying

$$\mathcal{F}_{n+1}/\ell^n\mathcal{F}_{n+1} \cong \mathcal{F}_n,$$

Thus we obtain the integral lisse ℓ-adic sheaf $\mathcal{F}$. Conversely, for a lisse sheaf $\mathcal{F}$, the stalk $\mathcal{F}_{u_0}$ has a natural $\pi_1^{\text{ét}}(U, u_0)$-action by taking the $\pi_1^{\text{ét}}(U, u_0)$-action with the induce map on each level, and assembling them together.

2.2.17. Let $\Lambda = \mathbb{Q}_\ell$. Given a continuous representation $\rho : \pi_1^{\text{ét}}(U, u_0) \to \mathrm{Aut}_{\mathbb{Q}_\ell}(M)$ on a finite-dimensional $\mathbb{Q}_\ell$-vector space M, choose a ρ-stable $\mathbb{Z}_\ell$-lattice $L \subset M$, which exists because the image of ρ is compact [Ser92, Part II, Ch. IV, Appendix 1, Theorem 1]. Apply 2.2.16 to the action on L, obtaining an integral lisse sheaf $\mathcal{F}_L$, and set $\mathcal{F}_\rho := \mathcal{F}_L \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. Conversely, choose an integral lisse model $\mathcal{G}$ of a rational lisse sheaf $\mathcal{F} \cong \mathcal{G} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. Extend the stalk action from 2.2.16 by scalars:

$$\mathcal{F}_{u_0} = \mathcal{G}_{u_0} \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell, \qquad \rho_{\mathcal{F}}(\gamma) := \rho_{\mathcal{G}}(\gamma) \otimes \mathrm{id}_{\mathbb{Q}_\ell}.$$

2.2.18 Example. (Local Systems from Torsors) Let $\Lambda : \mathsf{Ring}^{\mathrm{top}}$ be one of $\mathbb{Z}_\ell, \mathbb{Q}_\ell$, or $\mathbb{Z}/\ell^n\mathbb{Z}$ for $n : \mathbb{N}$. Given a finite group $G : \mathsf{Grp}^{\mathrm{fin}}$, a finitely generated Λ-module $M : \mathsf{Mod}_\Lambda^{\mathrm{top}}$, a continuous representation $\rho : G \to \mathrm{Aut}_\Lambda(M)$ with G given the discrete topology, and a finite étale right G-torsor $f : Y \to U$, we construct a local system $\mathcal{L}_\rho : \mathsf{LocSys}_\Lambda(U)$ by following the same procedure in 2.2.15, 2.2.16, and 2.2.17. In other words, the following.

1. For $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, as in 2.2.15, form the cover

$$E_\rho := Y \times^G M = (Y \times_U M_U)/G \xrightarrow{p} U,$$

   where $M_U := (\coprod_{m : M} U \to U) : \mathsf{Sch}_U$ and G acts by $(y, m) \cdot g = (yg, \rho(g)^{-1}m)$, then take the sheaf of sections to obtain the lisse sheaf

$$\mathcal{L}_\rho := \mathrm{Hom}_U(-, E_\rho) : \mathsf{LocSys}_\Lambda(U).$$

2. For $\Lambda = \mathbb{Z}_\ell$, as in 2.2.16, apply the finite-coefficient construction to $M_n := M/\ell^n M$ with its induced G-action to obtain $\mathcal{L}_n$. The G-equivariant reductions $M_{n+1} \twoheadrightarrow M_n$ induce maps of associated covers and hence transition maps $r_n : \mathcal{L}_{n+1} \to \mathcal{L}_n$ by composition of sections. These satisfy $\mathcal{L}_{n+1}/\ell^n\mathcal{L}_{n+1} \cong \mathcal{L}_n$, giving $\mathcal{L}_\rho : \mathsf{LocSys}_{\mathbb{Z}_\ell}(U)$.

3. For $\Lambda = \mathbb{Q}_\ell$, as in 2.2.17, choose a G-stable $\mathbb{Z}_\ell$-lattice $L \subset M$ and apply the integral construction L to obtain $\mathcal{L}_L$, then rationalize: $\mathcal{L}_\rho := \mathcal{L}_L \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell : \mathsf{LocSys}_{\mathbb{Q}_\ell}(U)$.

2.2.19 Example. We give some concrete implementations of 2.2.18.

1. Kummer
2. Artin–Shreier
3. Lang sheaves

2.2.20 Example. Tate twists

2.2.21 Example. smooth proper family, Kloostermann

2.3 The Tannakian Category Perspective

3 Perverse Sheaves and Middle Convolution

3.1 Constructible Sheaves

3.1.1 Notation. Define the collection of notations:

$X : \mathsf{Curv}_{\mathbb{C}}^{\mathrm{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \mathsf{Set}$	A finite nonempty subset of closed points of X
$U : \mathsf{Curv}_{\mathbb{C}}^{\mathrm{sm}}$	The open subvariety $X \setminus S$
$j : U \to X$	The inclusion map

$m : \mathbb{N}$	The cardinality of S
$(-)^{\mathrm{an}} : \mathsf{Curv}_{\mathbb{C}}^{\mathrm{sm}} \to \mathsf{Riem}$	The complex analytification functor

3.1.2 Definition. (Algebraically Constructible Sheaves)

3.1.3 Theorem. (Euler–Poincaré)

3.2 Six-Functor Formalism

3.3 Middle Extension Sheaves

3.4 Middle Convolution

4 Mod-ℓ Monodromy of Hyperelliptic Curves

Appendix

A.1 Analytic Continuation

A.1.1. Definition of Analytic Continuation.

A.2 Fundamental Groups of Punctured Riemann Surfaces

A.2.1 Proposition. *Let $X : \mathsf{Riem}$ be compact of genus $g : \mathbb{N}$, let $S = \{s_1, \ldots, s_m\}$ be a set of m distinct points of X, and put $U = X \setminus S$. For a base point $u_0 : U$, we have*

$$\pi_1^{\mathrm{top}}(U, u_0) \cong \left\langle a_1, b_1, \ldots, a_g, b_g, \gamma_1, \ldots, \gamma_m \;\middle|\; \prod_{i=1}^{g} [a_i, b_i]\, \gamma_1 \cdots \gamma_m = 1 \right\rangle,$$

where $[a, b] := aba^{-1}b^{-1}$, and the generators are:

1. *the loops running around the i-th handle $a_1, \ldots, a_g$;*
2. *the loops passing through the i-th handle $b_1, \ldots, b_g$;*
3. *the loops $\gamma_1, \ldots, \gamma_m$ encircling the puncture $s_1, \ldots, s_m$ once counterclockwise.*

Proof. See [Sza09, Theorem 3.6.3 and Remark 3.6.4]. □

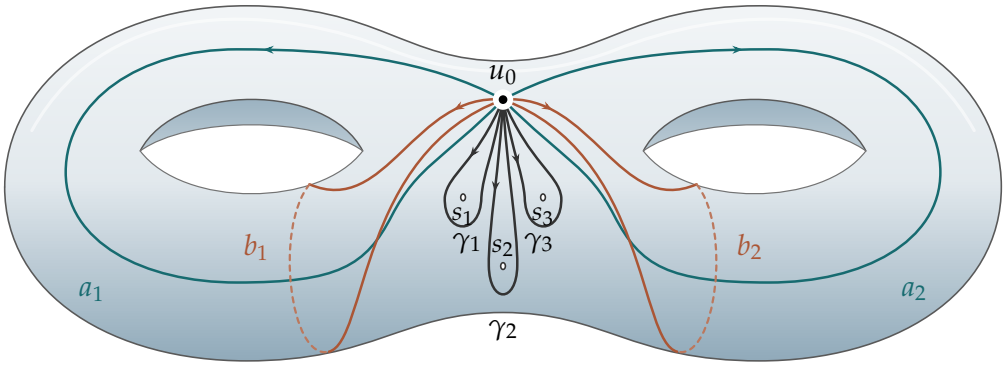


Figure 1: Figure for the case $g = 2$, $m = 3$, generated by GPT-6 Astra.

A.3 Grothendieck's Galois Theory

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